9

Rank Correlation Methods

Rank correlation coefficients are statistical indices that measure the degree of association between two variables having ordered categories. Some well-known rank correlation coefficients are those proposed by Goodman and Kruskal (1954, 1959), Kendall (1955), and Somers (1962). Rank correlation methods share several common features. They are based on counts and are defined such that a coefficient of zero means "no association" between the variables and a value of $+1.0$ or -1.0 means "perfect agreement" or "perfect inverse agreement," respectively.

Rank correlation methods have been popular in social science applications because of their analogy to Pearson product-moment correlation coefficients. They have had limited use in multivariate applications, however, because of the difficulty of computing tests of significance and using them for multivariate analysis. In the following pages we illustrate the use of rank correlation coefficients and demonstrate that these statistics are one example of a large class of complex functions that may be defined from the cell frequencies in the context of WLS theory.

ARE PEOPLE CONSISTENT IN THEIR RESPONSES?

Many urban areas in the United States face problems created by rapid growth. Houston, Texas, is one of the more extreme examples. In the middle 1970s one study estimated that Houston was growing at the rate of 1000 new people each week—more than 50,000 each year (Stafford and Lehnen, 1975).

Such rapid growth creates major problems for the delivery of urban services and profoundly affects the city's quality of life. The capacity of organizations to provide education, sanitation, mass transportation, housing, and public health is often strained to the limit, if not overwhelmed, and the quality of life—measured by social indicators such as air quality and crime statistics—declines. Perceiving this all-too-familiar scenario, three Houston-based corporations commissioned a research project, the Houston Community Study, to measure among other things the attitudes and opinions of a cross-section of Houstonians toward the problems created by urban growth (Stafford and Lehnen, 1975). The data were to be used by business firms and social organizations based in the city for establishing priorities among community service projects.

The sponsors of the project were concerned that public attitudes toward policy issues could not be measured reliably. Previous opinion research conducted by the University of Michigan Survey Research Center reported that some sectors of the public lacked sufficient knowledge and attentiveness to form stable opinions about broad questions of policy (Converse, 1964). This conclusion was arrived at by comparing responses among a set of respondents to successive administrations of the same question. The varying patterns of consistency and inconsistency suggested that certain issues, especially questions of domestic policy, were beyond the attention range of respondents with lower social status. In response to these considerations, the study team developed a methodological study as part of a household sample survey to measure the consistency of response among various groups of individuals and to test the effect of education on response consistency.

Fifteen sets of policy statements were administered by means of a card-sort technique. At two points during the interview respondents were given a deck of about 50 cards containing statements eliciting an agree/disagree response. Respondents were asked to sort the cards into boxes labeled "agree," "disagree," "have no opinion," and "not sure." Since identical repetitions of policy statements were included in each card deck, the respondents replied twice to the same statement during the course of the interview.

Table 9.1 reports response data for two administrations of the statement measuring attitudes toward integration of the schools by level of the respondent's education. Each subtable is size (4×4), representing the 16 combinations of two responses to the question "The government should make sure that white and Negro children go to the same school together."

Why is response consistency so low? One prevailing theory argues that respondents, desiring to "please," to "be helpful," or to hide their lack of understanding, respond to a question even when they have no idea of its meaning. To reduce this bias, interviewers instructed the respondents in the difference

Table 9.1 Responses to the Statement "The Government Should Make Sure That White and Negro Children Go to the Same School Together"

(a) Respondents with Less Than High School Education ($n = 252$)

	SECOND RESPONSE			
FIRST RESPONSE	*Disagree*	*Not Sure*	*Agree*	*Have No Opinion*
Disagree	86	4	13	3
Not Sure	5	5	8	3
Agree	11	5	74	4
Have no opinion	7	2	7	15

(b) Respondents with High School Diploma ($n = 235$)

	SECOND RESPONSE			
FIRST RESPONSE	*Disagree*	*Not Sure*	*Agree*	*Have No Opinion*
Disagree	86	5	13	2
Not Sure	5	0	8	2
Agree	16	1	61	6
Have no opinion	4	4	12	10

(c) Respondents with Some College Education ($n = 335$)

	SECOND RESPONSE			
FIRST RESPONSE	*Disagree*	*Not Sure*	*Agree*	*Have No Opinion*
Disagree	144	5	25	4
Not Sure	7	23	10	1
Agree	14	4	67	5
Have no opinion	5	4	3	14

Table 9.2 Presence or Absence of Opinion by Education

(*a*) *Less Than High School* ($n = 252$)			(*b*) *High School* ($n = 235$)			(*c*) *Some College* ($n = 335$)		
	0	*No*		*0*	*No*		*0*	*No*
0	211	10	0	195	10	0	299	10
No	16	15	No	20	10	No	12	14

0 = expressed agree/not sure/disagree response; No = "have no opinion" response.

between "not having an opinion" and being ambivalent ("not sure"). Thus the "not sure" box was placed in the middle of the agree/disagree continuum and the "have no opinion" box was placed to one side of the card-sort board. The data arrayed in Table 9.1 reflect this distinction; the respondents who indicated opinions on two occasions are grouped in the 3×3 area in the upper left of each subtable.

Two different patterns of consistency are of interest here. One is for the subset of each education level who expressed agree/not sure/disagree opinions twice. The second pattern analyzed is the opinion versus "have no opinion" comparison. Each subtable in Table 9.1 can be reduced to 2×2 tables as shown in Table 9.2. These three subtables are used to analyze one function; the 3×3 subtables for people expressing an opinion twice are used to construct a second function.

A RANK CORRELATION COEFFICIENT

Although data such as those reported in Table 9.1 present many alternatives for statistical analysis, we will demonstrate the approach common to the sociological literature (Converse, 1964). This approach uses rank correlation measures, such as Goodman and Kruskal's gamma, to measure the degree of consistency between two responses. The *gamma statistic* is a convenient choice because it has a possible range of -1.0 to $+1.0$, and it may be interpreted as the *proportion of untied pairs* of respondents that *agree* minus the proportion that *disagree*. Gamma measures only one type of response consistency: the tendency of the responses to group along the major or minor diagonals of a table.

The following two examples demonstrate that gamma varies according to the distribution of the data in the body of the table even though the marginal distributions remain the same. Note that the body of the tables shows considerable differences in *gross* change, to which the gamma statistics are sensitive, although there are no differences in the marginal distributions:

Example of Frequency Table
Having Low Gamma ($\hat{G} = -0.17$)

RESPONSE TO FIRST QUESTION	RESPONSE TO SECOND QUESTION *Agree*	*Disagree*	*Total*
Agree	20	28	48
Disagree	20	20	40
Total	40	48	88

Example of Frequency Table
Having High Negative Gamma ($\hat{G} = -0.98$)

RESPONSE TO FIRST QUESTION	RESPONSE TO SECOND QUESTION *Agree*	*Disagree*	*Total*
Agree	4	44	48
Disagree	36	4	40
Total	40	48	88

Rank correlation coefficients such as gamma, Kendall's tau, and Somers' d_{yx} and d_{xy} are some of the more complicated functions that may be analyzed in the context of the WLS approach (Forthofer and Koch, 1973). Rank correlation coefficients require the computation of the following quanitities:

$$A = \text{number of pairs in agreement}$$

$$D = \text{number of pairs in disagreement}$$

$$T_x = \text{number of pairs tied on variable } X$$

$$T_y = \text{number of pairs tied on variable } Y$$

$$T_{xy} = \text{number of pairs tied on both variables}$$

$$n = \text{sample size}$$

$$\text{TOT} = \text{total number of pairs} = A + D + T_x + T_y + T_{xy} = n(n-1)/2$$

The estimates of Goodman and Kruskal's gamma (G), Kendall's tau (T), and Somers' measures (d_{yx} and d_{xy}) use the difference $A - D$ in the numerator. The estimate of gamma (G) is defined as

$$G = \frac{A - D}{A + D}$$

where the denominator is the number of untied pairs only. Kendall's tau is estimated by

$$T = \frac{A - D}{\sqrt{(A + D + T_x)(A + D + T_y)}}$$

Estimates of Somers' asymmetric measures d_{yx} and d_{xy} are defined as

$$d_{yx} = \frac{A - D}{A + D + T_y}$$

$$d_{xy} = \frac{A - D}{A + D + T_x}$$

In general, the various rank correlation coefficients differ in their treatment of the tied pair counts T_x, T_y, and T_{xy} in the denominator. Each function may be analyzed by using the WLS approach.

Table 9.3 summarizes the algebraic computations for rank correlations for the agree/not sure/disagree and the opinion/no opinion patterns for data taken from panel (*a*) of Table 9.1. The example illustrates that all operations are based on the 16 cell counts $n_{11}, n_{12}, \ldots, n_{1,16}$. We obtain the count of concordant pairs (A) by starting with the upper left-hand cell frequency and multiplying it by every cell count that is below and to the right—that is, $n_{11}(n_{16} + n_{17} + n_{1,10} + n_{1,11})$. The cells below and to the right have more positive feelings (not sure and agree are more positive than disagree) on *both* administrations of the question than the cell that contains people who said disagree both times. We take every cell in the table with observations below and to the right—namely, cells n_{11}, n_{12}, n_{15}, and n_{16}—and perform a similar operation. Summing these counts produces A.

To determine the number of discordant pairs (D), we use a similar rule except that we start in the upper right-hand corner and add the products of cells below and to the left. Anderson and Zelditch (1968) provide an easy-to-read description of the computational steps for deriving the counts for A, D, T_x, T_y, T_{xy}, and TOT and calculating the values of several rank correlation coefficients.

Table 9.3 Computing Gamma Coefficients for Data Reported in Table 9.1 (Panel *a*)

(a) Frequencies n_{ij}

Variable 1	Variable 2: Low ←		→ High	No Opinion
Low	$n_{11} = 86$	$n_{12} = 4$	$n_{13} = 13$	$n_{14} = 3$
↕	$n_{15} = 5$	$n_{16} = 5$	$n_{17} = 8$	$n_{18} = 3$
High	$n_{19} = 11$	$n_{1,10} = 5$	$n_{1,11} = 74$	$n_{1,12} = 4$
No opinion	$n_{1,13} = 7$	$n_{1,14} = 2$	$n_{1,15} = 7$	$n_{1,16} = 15$

(b) Computing Gamma for Consistency of Response (among Ordered Categories)

Number of Pairs in Agreement (A)

$$n_{11}(n_{16} + n_{17} + n_{1,10} + n_{1,11}) = 86(5 + 8 + 5 + 74) = 7912$$
$$n_{12}(n_{17} + n_{1,11}) = 4(8 + 74) = 328$$
$$n_{15}(n_{1,10} + n_{1,11}) = 5(5 + 74) = 395$$
$$n_{16}(n_{1,11}) = 5(74) = 370$$
$$A = 9005$$

Number of Pairs in Disagreement (D)

$$n_{13}(n_{15} + n_{16} + n_{19} + n_{1,10}) = 13(5 + 5 + 11 + 5) = 338$$
$$n_{12}(n_{15} + n_{19}) = 4(5 + 11) = 64$$
$$n_{17}(n_{19} + n_{1,10}) = 8(11 + 5) = 128$$
$$n_{16}(n_{19}) = 5(11) = 55$$
$$D = 585$$

$$\text{Gamma}(G) = \frac{A - D}{A + D} = \frac{9005 - 585}{9005 + 585} = 0.88.$$

(c) Computing Gamma for Consistency of Response (Opinion vs. No Opinion)

Number of Pairs in Agreement (A)

$$(n_{11} + n_{12} + n_{13} + n_{15} + n_{16} + n_{17} + n_{19} + n_{1,10} + n_{1,11})n_{1,16} = 211(15) = 3165$$
$$A = 3165$$

Number of Pairs in Disagreement (D)

$$(n_{14} + n_{18} + n_{1,12})(n_{1,13} + n_{1,14} + n_{1,15}) = 10(16) = 160$$
$$D = 160$$

$$\text{Gamma}(G) = \frac{A - D}{A + D} = \frac{3165 - 160}{3165 + 160} = 0.90.$$

COMPUTING GAMMA USING GENCAT

We demonstrate the creation of the two gammas shown in Table 9.3 by using the matrix operations found in the GENCAT program. Although there are more efficient methods for computing the correlation coefficients and their variance-covariance estimates, this example shows how a complex function is produced by a combination of linear, logarithmic, and exponential transformation of the cell counts.

The data for a single population are arranged as in Table 9.1. The vector of observed probabilities is

$$\underset{(16\times 1)}{\mathbf{p}_1} = \begin{bmatrix} p_{11} \\ p_{12} \\ \vdots \\ p_{1,16} \end{bmatrix}$$

The calculations shown in Table 9.3 indicate that the following 20 quantities are required to find the number of concordant and discordant pairs:

Concordant	*Discordant*
Opinion vs. No Opinion	Opinion vs. No Opinion
$(n_{11}+n_{12}+n_{13}+n_{15}+n_{16}+n_{17}+n_{19}+n_{1,10}+n_{1,11})$	$(n_{14}+n_{18}+n_{1,12})$
$n_{1,16}$	$(n_{1,13}+n_{1,14}+n_{1,15})$
Agree/Disagree	Agree/Disagree
n_{11}	n_{13}
$(n_{16}+n_{17}+n_{1,10}+n_{1,11})$	$(n_{15}+n_{16}+n_{19}+n_{1,10})$
n_{12}	n_{12}
$(n_{17}+n_{1,11})$	$(n_{15}+n_{19})$
n_{15}	n_{17}
$(n_{1,10}+n_{1,11})$	$(n_{19}+n_{1,10})$
n_{16}	n_{16}
$n_{1,11}$	n_{19}

Each of these 20 quantities is an individual cell count or the sum of the counts. We can create the sums and quantities listed above by forming linear combinations of the probabilities—that is, by premultiplying $\mathbf{p}_1$ by the matrix $\mathbf{A}_1$:

$$\underset{(20\times 16)}{\mathbf{A}_1} = \begin{bmatrix} \underset{(2\times 16)}{\mathbf{A}_{11}} \\ \underset{(2\times 16)}{\mathbf{A}_{12}} \\ \underset{(8\times 16)}{\mathbf{A}_{13}} \\ \underset{(8\times 16)}{\mathbf{A}_{14}} \end{bmatrix}$$

where

Opinion vs. No Opinion, Concordant

$$\underset{(2\times 16)}{\mathbf{A}_{11}} = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Opinion vs. No Opinion, Discordant

$$\underset{(2\times 16)}{\mathbf{A}_{12}} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Agree/Disagree, Concordant

$$\underset{(8\times 16)}{\mathbf{A}_{13}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Agree/Disagree, Discordant

$$\underset{(8\times 16)}{\mathbf{A}_{14}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The next step is to create the products indicated in Table 9.3. Since we know that $\ln(a) + \ln(b) = \ln(ab)$, we may add logarithms of the appropriate terms. Taking the logarithms of the 20 terms formed by $\mathbf{A}_1\mathbf{p}_1$ yields $\ln(\mathbf{A}_1\mathbf{p}_1)$; adding the appropriate pairs of logarithms produces the logarithms of the desired products. The $\mathbf{A}_2$ matrix that forms the sums of the appropriate pairs is

$$\underset{(10\times 20)}{\mathbf{A}_2} = \begin{bmatrix} 1&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&1&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&1&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&1&1&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&1&1&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&1&1&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&1&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&1&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&1&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&1 \end{bmatrix}$$

$$\underset{(10\times 20)}{\mathbf{A}_2} = \begin{bmatrix} \underset{(1\times 2)}{\mathbf{A}_2^*} & & & \\ & \mathbf{A}_2^* & . & \\ & & . & \\ & & \cdot & \mathbf{A}_2^* \end{bmatrix}$$

where $\mathbf{A}_2^* = [1 \quad 1]$. The diagonal structure of $\mathbf{A}_2$ results from the $\mathbf{A}_1$ matrix, which puts terms to be multiplied adjacent to one another.

If we take the exponential of the logarithmic functions, we now have the desired products. The next step is to form $A - D$ and $A + D$. This is done by summing the appropriate products as indicated in Table 9.3. The $\mathbf{A}_3$ matrix performs this summation:

$$\underset{(4\times 10)}{\mathbf{A}_3} = \begin{bmatrix} 1&-1&0&0&0&0&0&0&0&0\\ 1&1&0&0&0&0&0&0&0&0\\ 0&0&1&1&1&1&-1&-1&-1&-1\\ 0&0&1&1&1&1&1&1&1&1 \end{bmatrix} \begin{array}{l} \textit{row label} \\ A - D \text{ (opinion/no opinion)} \\ A + D \text{ (opinion/no opinion)} \\ A - D \text{ (agree/disagree)} \\ A + D \text{ (agree/disagree)} \end{array}$$

We now have formed the numerator and denominators of the gammas. To take their ratio, we again use logarithms because $\ln a - \ln b = \ln(a/b)$. Premultiply $\ln\{\mathbf{A}_3 \exp[\mathbf{A}_2 \ln(\mathbf{A}_1\mathbf{p}_1)]\}$ by $\mathbf{A}_4$, where

$$\underset{(2\times 4)}{\mathbf{A}_4} = \begin{bmatrix} 1&-1&0&0\\ 0&0&1&-1 \end{bmatrix}$$

We now have $\ln[(A - D)/(A + D)]$. To obtain $(A - D)/(A + D)$, take the exponential of these functions, which produces the desired gammas. The complex function in matrix terms is

$$\underset{(2\times 1)}{\mathbf{F}} = \exp(\mathbf{A}_4 \ln\{\mathbf{A}_3 \exp[\mathbf{A}_2 \ln(\mathbf{A}_1\mathbf{p})]\}) = \begin{bmatrix} 0.88 \\ 0.90 \end{bmatrix}$$

The value $G = 0.88$ computed for the data in panel (*a*) of Table 9.1 is a "partial gamma" in the sense that it is the correlation of response 1 and response 2 *only* for individuals with less than a high school education who twice expressed an opinion about the school integration statement. Since there are three subpopulations defined by education level and two coefficients computed for each subpopulation, there are six estimated gamma coefficients of interest in this problem. These six coefficients, computed by the GENCAT program, are shown in Table 9.4. Because the quantities A, D, T_x, T_y, and T_{xy} can be formed by linear, logarithmic, and exponential operations, analyses using functions such as gamma, tau, or Somers' measures are within the scope of WLS methodology.

Table 9.4 Gamma Coefficients for Data in Table 9.1

Education	*Gamma for Expressing Opinion vs. No Opinion*	*Gamma for Two Agree/ Not Sure/Disagree Responses*
Less than high school	0.90	0.88
High school	0.81	0.85
Some college	0.94	0.81

Two problems sometimes arise when one is using matrix manipulation in the GENCAT program to compute complex functions from frequency data. First, the number of functions computed in the intermediate steps to derive the rank correlations is large and can often exceed the storage capacities of many computers. The second problem arises from the fact that it is sometimes necessary to take the natural logarithm of a cell count of zero, but the natural logarithm of zero is undefined. This problem arises in the computation of the gammas from panel (*b*) of Table 9.1. The recommended solution is to substitute a nonzero value—say $1/r$, where r is the number of response categories for the subpopulation. (See Chapter 2 for further discussion of this point.) For this problem $r = 16$ and $1/r = 0.06$. The practical effect on the statistical estimates is negligible: The exact values of the estimated gamma coefficients are 0.85386 and 0.81395, whereas the values computed by GENCAT using the $1/r$ substitution rule are 0.85338 and 0.81401. Analysts may avoid both problems by writing a special computer routine or using software available at a local computer center to calculate **F** and $\mathbf{V}_F$, which may then be input directly to the GENCAT program (see Appendix D).

EXAMINING CONSISTENCY

The purpose of computing the six estimated gamma coefficients is to examine the relationship between education and response consistency. The sociological literature suggests that respondents with little education will be least consistent and highly educated respondents will be most consistent (Converse, 1964). Since we are measuring two types of consistency, we also expect a difference between the opinion/no opinion and the agree/disagree coefficients. We begin our investigation with the model

$$\mathbf{X}_1\boldsymbol{\beta}_1 = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \beta_{10} \\ \beta_{20} \\ \beta_{11} \\ \beta_{21} \end{bmatrix}$$

where β_{10} = mean gamma for agree/disagree response
β_{20} = mean gamma for opinion/no opinion response
β_{11} = coefficient of a linear education term (we have assumed equally spaced levels of education) for agree/disagree response
β_{21} = coefficient of a linear education term for opinion/no opinion response

The linear education term is shown in the **X** matrix by -1 if the group has less than a high school education, by 0 if the group has graduated from high school but not attended college, and by 1 if the group has some college education. Note that other values could be assigned here if one does not wish to assume equal spacing for the educational categories.

The test statistic for the fit of this model is $X^2_{\mathrm{GOF}} = 1.55$. This value is less than the seventy-fifth percentile of the chi-square distribution ($\chi^2_{2,0.75} = 2.77$)—a preliminary indication of an adequate fit.

The parameter estimates are

$$\mathbf{b} = \begin{bmatrix} 0.85 \\ 0.91 \\ -0.03 \\ 0.02 \end{bmatrix}$$

The parameter estimates b_{11} and b_{12} suggest that an inverse relationship exists between education and consistency (gamma) for the agree/disagree response ($b_{11} = -0.03$) and a direct relationship between education and consistency for the opinion/no opinion response ($b_{21} = 0.02$). Tests reveal that neither of these effects is significantly different from zero, however. The test statistics for the hypotheses of no linear education effects are $X_1^2 = 1.67$ and $X_1^2 = 0.89$ for the agree/disagree response and the opinion/no opinion response, respectively. These are both less than $\chi^2_{1,0.95} = 3.84$, so we may comfortably delete these terms from the model. The test statistic for the hypothesis that the mean gammas are not significantly different from one another (H_0: $\beta_{10} = \beta_{20}$) is $X_1^2 = 4.15$. This is greater than the ninety-fifth percentile of the chi-square distribution and hence we reject the hypothesis of no difference.

Because there are no statistically significant education effects, we fit a reduced model that includes β_{10} and β_{20} only. Our original estimates of β_{10} and β_{20} are modified slightly by the presence of the education terms; therefore, to obtain clean estimates not contaminated by the nonsignificant education effects, we fit this reduced model. The model fits well: The test statistic for lack of fit is $X_4^2 = 4.11$, which is less than $\chi^2_{4,0.75} = 5.39$. The estimates of the mean gammas are $b_{10} = 0.85$ and $b_{20} = 0.92$. These two values differ statistically as the test statistic of the hypothesis of no difference is $X_1^2 = 5.63$ and $\chi^2_{1,0.95} = 3.84$. Hence the consistency level differs between the agree/disagree response and the opinion/ no opinion response.

RESULTS: NO EDUCATION EFFECT

The data on responses to the school integration question do not support the expectation that people with lower education are less consistent. Both measures of consistency—one measuring the consistency in forming an opinion and the other measuring stability of opinion—show that education had no statistically significant effect on these two measures of association. The estimated gamma coefficient for the formation of an opinion (opinion versus no opinion) is 0.92 and the estimated gamma for opinion stability is 0.85. The data suggest a slight, but not statistically significant, linear effect of education level on the estimated gamma coefficient measuring agree/disagree opinion stability, but the estimated effect is not in the hypothesized direction. Moreover, inspection of the observed gammas for the opinion/no opinion coefficients suggests a possible quadratic relationship between education and consistency. We did not test the result with a strict test, because we had not hypothesized the relationship beforehand.

SUMMARY

This chapter has demonstrated the feasibility of computing measures of association such as the rank correlation coefficient gamma with the WLS methodology. These statistical measures have a wide range of application, especially in the social sciences, but the limitations of space prevent the presentation of further examples.

The key topics covered in this chapter are:

- *The problem:* Determine whether consistency of response to survey questions varies by education level.
- *Rank correlation:* A method of analyzing multiple responses; we used gamma coefficient in this chapter.
- *Consistency:* All the gammas had large values, at least 0.80.
- *Results:* The consistency measures were not affected by education in a linear fashion.

Chapters 8 and 9 have shown three ways of examining multiresponse data. In Chapter 10 we use the mean score approach to a different multiresponse situation: the assignment of ranks.

EXERCISES

9.1.

Analyze the data presented in Exercise 8.1 using the gamma function.

a. Compute the estimated gamma function and the associated variance-covariance matrix for each subtable. Test the hypothesis that there is no linear effect of education on response consistency (as measured by gamma) between the first and second response.

b. Analyze these data using the function "proportion consistent from time 1 to time 2." Test the hypothesis that consistency is not affected by education.

c. Construct a *net change score* for these data that is similar to the function defined in Exercise 6.1. Do you consider this function suitable for this application? Explain your answer. Test the following hypotheses using this function:

1. The net change score is not affected by level of education.
2. The net change score is zero for each level of education.

d. Compare your findings to the analyses performed previously. Which analysis seems best for these data? Explain.

9.2.

The data in Table 9.5 are from a national ulcer study. The response variable is the severity of the *dumping syndrome*, a possible undesirable consequence of surgery for duodenal ulcer. Four hospitals and four surgical procedures are

Table 9.5 Dumping Syndrome by Operation Type and Hospital

	OPERATION *A*			OPERATION *B*			OPERATION *C*			OPERATION *D*		
HOSPITAL	*None*	*Slight*	*Mod.*	*None*	*Slight*	*Mod.*	*None*	*Slight*	*Mod.*	*None*	*Slight*	*Moc*
I	23	7	2	23	10	5	20	13	5	24	10	6
II	18	6	1	18	6	2	13	13	2	9	15	2
III	8	6	3	12	4	4	11	6	2	7	7	4
IV	12	9	1	15	3	2	14	8	3	13	6	4

Source: Grizzle, Starmer, and Koch (1969).

reported in this problem. The surgical procedures are

A: drainage and vagotomy

B: 25 percent resection (antrectomy) and vagotomy

C: 50 percent resection (hemigastrectomy) and vagotomy

D: 75 percent resection

Procedure *A* is least radical and procedure *D* is most radical.

a. Form a measure of association between the severity of the dumping syndrome and the surgical procedure for each of the four hospitals.

b. Does the measure of association depend on the hospital variable? How would you interpret these findings?

c. What recommendations, if any, would you make to the administrators of the hospitals? What reservations would you place on your recommendations? Why?

d. Use the mean score approach from Chapter 6 and examine the relationship between a mean severity score and the hospital and type-of-operation variables.

e. Compare the results of both analyses.

9.3.

In Exercise 7.4 the log cross-product ratio function was used to characterize the association between attitudes toward the candidate and voting preferences at two points in time. Another function characterizing the association is Goodman and Kruskal's gamma (G). Use the estimated gamma function to test whether the association between attitude toward the Republican candidate and voting preference is the same for both time periods.

a. Construct the following functions using GENCAT:

G_{vc1} = rank correlation coefficient for voting preference and attitude toward candidate at time 1

G_{vc2} = rank correlation coefficient for voting preference and attitude toward candidate at time 2

These functions and their variance-covariance structure may be estimated by using GENCAT, other software, or hand calculation. The following steps illustrate the matrix approach used in the GENCAT program:

1. $\mathbf{A}_1\mathbf{p}$ produces $\mathbf{F}_1$.
2. $\ln(\mathbf{F}_1)$ produces $\mathbf{F}_2$.
3. $\mathbf{A}_2\mathbf{F}_2$ produces $\mathbf{F}_3$.
4. $\exp(\mathbf{F}_3)$ produces $\mathbf{F}_4$.
5. $\mathbf{A}_3\mathbf{F}_4$ produces $\mathbf{F}_5$.
6. $\ln(\mathbf{F}_5)$ produces $\mathbf{F}_6$.
7. $\mathbf{A}_4\mathbf{F}_6$ produces $\mathbf{F}_7$.
8. $\exp(\mathbf{F}_7)$ produces $\mathbf{F}_8$.

The $\mathbf{p}$ vector is the same as in Exercise (7.4).

$$\mathbf{A}_1 = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} \textit{row label} \\ a\text{: R+, time 1} \\ b\text{: R}-\text{, time 1} \\ c\text{: D+, time 1} \\ d\text{: D}-\text{, time 1} \\ a\text{: R+, time 2} \\ b\text{: R}-\text{, time 2} \\ c\text{: D+, time 2} \\ d\text{: D}-\text{, time 2} \end{array}$$

$\mathbf{A}_2$ is a 4×8 matrix; $\mathbf{A}_3$ is a 4×4 matrix; $\mathbf{A}_4$ is a 2×4 matrix.

$$\mathbf{F}_8 = \begin{bmatrix} G_{vc1} \\ G_{vc2} \end{bmatrix}$$

b. Test the following hypotheses:

1. The association between attitude toward the Republican candidate and voting preference at time 1 is zero.

2. The association between attitude toward the Republican candidate and voting preference at time 2 is zero.
3. The association between attitude toward the Republican candidate and voting preference is constant across time periods.

Explain the relationship between tests (1 and 2) and (3) above. If one cannot reject the hypothesis of no association for tests (1) and (2), is it necessary to test (3)?

APPENDIX: GENCAT INPUT

The following listing is for the GENCAT control cards that reproduce the analysis described in this chapter.

```
    5     1     1                     CHAPTER 9: RANK CORRELATION ANALYSIS
    3    16                           (16F3.0)
 86  4 13  3  5  5  8  3 11  5 74  4  7  2  7 15
 86  5 13  2  5.06  8  2 16  1 61  6  4  4 12 10
144  5 25  4  7 23 10  1 14  4 67  5  5  4  3 14
    1     2    60    20     1         (20F2.0)
 1 1 1 0 1 1 1 0 1 1 1 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
 0 0 0 1 0 0 0 1 0 0 0 1 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 0
 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 1 1 0 0 1 1 0 0 0 0 0
 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0
 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0
 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 1 1 0 0 1 1 0 0 0 0 0 0
 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0
 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0
 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0
    2
    1     2    30    10               (20F2.0)
 1 1
000001 1
0000000001 1
00000000000001 1
000000000000000001 1
0000000000000000000001 1
00000000000000000000000001 1
000000000000000000000000000001 1
000000000000000000000000000000000101
0000000000000000000000000000000000000101
    3
    1     2    12     4               (10F2.0)
01-1
0101
```

```
000001010101-1-1-1-1
000001010101010101010101
     2
     1     2     6     2                (4F2.0)
01-1
000001-1
     3
     7     1     4                      (6F2.0)
010101010101
01-101-101-1
-10000000100
00-100000001
     8     1     1                       (4F2.0)
0001
     8     1     1                       (4F2.0)
000001
     8     1     1                       (4F2.0)
00000001
     7     1     3                        (6F2.0)
010101010101
01-101-101-1
00-100000001
     8     1     1                       (3F2.0)
000100
     8     1     1                       (3F2.0)
000001
     7     1     2                        (6F2.0)
010101010101
01-101-101-1
     8     1     1                       (3F2.0)
000100
     7     1     1                        (6F2.0)
010101010101
```